Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Syntax 404
Project / Solution Name	Faraway — Career and Education Compass
Name (Leader)	LOW XIN HUI
Email & Phone (Leader)	SWE2509070@xmu.edu.my +60 18-977 3221
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://faraway-practice-faraway.vercel.app/login
Code Repository URL (public / access granted to judges)	https://github.com/xxin73/SDCHACKATHON_SYNTAX404_FARWAY.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Faraway is a web application for individual career preparation, HR workforce development, and university or TVET student development. Its React and TypeScript interface supports profiles, skill assessments, career comparisons, learning plans, and progress tracking. Supabase provides email authentication, PostgreSQL storage for personal profiles and assessments, and Edge Functions for authenticated AI requests. Application code calculates career-match scores, skill gaps, and moderate or rapid AI-scenario comparisons. Qwen3-4B-Instruct-2507, hosted through RunPod Serverless and vLLM, generates structured explanations and learning plans from supplied context.

The AI request flow is: **user interface → Supabase authentication and Edge Function → RunPod-hosted Qwen model → response validation → results displayed to users**. Database operations occur separately as needed; the database is not simply a stage after AI inference. Organisation workspaces currently use account-scoped browser storage. O*NET occupational information, DOSM Malaysian labour-market statistics, and BLS US employment projections support analysis with source dates and geographical limitations. GitHub Actions schedules supported evidence updates.

Replit supports development and testing, GitHub maintains source code, and Vercel hosts the website. Server-side secrets protect AI credentials.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	Licence
TypeScript 5.9.x	Typed application and backend logic	Apache 2.0
JavaScript / HTML / CSS	Browser behaviour, structure, and styling	Languages/standards; implementations and libraries have their own licences
React and React DOM 19.1.0	User interface and rendering	MIT
Vite ^7.3.2	Development server and production build	MIT
Tailwind CSS ^4.1.14	Responsive styling	MIT
Recharts ^2.15.2	Skill and job-trend charts	MIT
Wouter ^3.3.5	Frontend routing	ISC
Framer Motion ^12.23.24	Interface animation	MIT
Supabase JavaScript client ^2.118.0	Authentication and database access	MIT
TanStack React Query ^5.90.21	Frontend asynchronous data management	MIT
Radix UI components — multiple versions	Accessible interface components	MIT
React Hook Form ^7.55.0 and Zod ^3.25.76	Form handling and validation	MIT
Lucide React ^0.545.0	Interface icons	ISC
PDF.js / pdfjs-dist ^6.4.299	PDF text extraction	Apache 2.0
Tesseract.js ^7.0.0	OCR support	Apache 2.0
Python — version varies by environment	Evidence preparation and translation scripts	Python Software Foundation licence
PostgreSQL / PL/pgSQL — managed Supabase version	Personal data storage and database functions	PostgreSQL licence
vLLM — deployed worker version to confirm	Serving the Qwen model on RunPod	Apache 2.0
CTranslate2 and SentencePiece — installed versions to confirm	CPU translation preparation	MIT / Apache 2.0

Additional direct and transitive dependencies are recorded in the repository's package manifests and lockfile. Their applicable licence and attribution notices are retained.

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier
RunPod inference API	Calls the hosted Qwen model for structured explanations and learning plans	Credit-funded GPU inference; credits were sponsor-provided
Supabase Auth, database and Edge Function APIs	Authentication, personal data operations, and backend AI requests	Free project tier, subject to account limits; confirm
Google Gemini API	Used during earlier development; optional provider support remains in the code (RunPod is the selected provider. Gemini support is not an automatic failover; switching providers requires server configuration.)	Free tier
BLS public web/data endpoints	Retrieves occupational employment projection evidence	Public access
DOSM/OpenDOSM public data endpoints	Retrieves Malaysian labour-market context	Public access
O*NET public release information	Checks occupational database release metadata	Public access; reviewed occupational data imports are separate
arXiv feed/API	Retrieves dated AI research context	Public access, subject to service usage policies
Google Fonts	Loads interface fonts	Public font-hosting service

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel	Hosts the production frontend and deployed API handlers	Hobby (Free); global CDN (static frontend)
Supabase	Authentication, PostgreSQL, personal data storage, and Edge Functions	Free tier; project database region: Northeast Asia, Tokyo
RunPod Serverless	GPU inference using Qwen and vLLM	Credit-funded; worker region depends on endpoint availability
Replit	Development environment, interface editing, preview, and testing	Sponsor-provided credits; region managed by the platform
GitHub / GitHub Actions	Source control and scheduled evidence-update workflow	GitHub Free plan; GitHub-hosted workflow runners

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage	Licence / Terms
Qwen3-4B-Instruct-2507	Qwen; hosted on RunPod	Pretrained; prompted for explanations and learning plans; no team fine-tuning	Apache 2.0
NLLB-200 distilled 600M	Meta; used through CPU translation tooling	Pretrained; prepares translated interface/catalogue/evidence text; not the live conversational model	CC BY-NC 4.0
Gemini — Gemini 3.5 Flash Lite	Google	Earlier prototype inference; optional provider implementation remains	Google API terms applicable to the model and account
Tesseract OCR language models	Tesseract project	Pretrained OCR for supported document extraction	Applicable trained-data licence; verify the files used

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type	Licence / Terms	Personal Data?
O*NET occupational and skill information, including releases 30.3 and 31.0	https://www.onetcenter.org/database.html	Public	CC BY 4.0, subject to stated exceptions	No
DOSM labour-market statistics and quarterly job-offering evidence	https://open.dosm.gov.my/ and the specific DOSM publications cited in charts	Public	Applicable dataset/publication terms; attribution provided	No
BLS occupational employment and projections	https://www.bls.gov/emp/	Public	Public-domain material, subject to stated exceptions; cite BLS	No
arXiv AI research metadata and linked publications	https://arxiv.org/	Public	Metadata/API terms and individual publication licences	Yes—public author names may appear
Selected ESCO occupational/skill references	https://esco.ec.europa.eu/	Public	Applicable European Commission reuse terms; attribution required	No
Personal profiles, skill assessments, and learning progress	Consenting application users	Private / self-collected	Purpose-limited application data; not an open dataset	Yes
HR employee and university student assessments	User-entered records or CSV imports	Private / self-collected; synthetic records for demonstrations	Permission-based use; not an open dataset	Yes for identifiable records

Dataset Name	Source / URL	Type	Licence / Terms	Personal Data?
Development test fixtures	Team-created sample profiles and records	Synthetic	Team-authored	No

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use
ChatGPT / Codex — service model versions vary	Code generation and revision, debugging, testing support, architecture, documentation, and proposal drafting	Major
Replit Agent — version managed by provider	Development assistance and troubleshooting	Major
Google Gemini / AI Studio — Gemini 3.5 Flash Lite	Earlier AI integration and testing	Minor
NLLB translation tooling	Preparation of translated interface and catalogue content	Major within translation preparation

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team reviewed AI-assisted code and content through source inspection, compilation checks, targeted logic tests, and manual interface walkthroughs. Testing identified issues in CSV imports, account isolation, programme filtering, and chart presentation, which informed revisions. We checked evidence descriptions against their sources and distinguished application scenarios from official forecasts. These checks do not establish that every output is correct or unbiased. The team remains responsible for the submitted code, documentation, source attribution, and remaining limitations. AI-generated guidance is not a guarantee of employment, professional eligibility, or learning outcomes.AI-generated content was reviewed and adjusted where necessary before being used in the project. Our team remains responsible for verifying the accuracy, quality, and appropriateness of all AI-generated outputs. AI-generated results are treated as guidance rather than guaranteed recommendations.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource	How It Was Used
Replit	Development platform and any confirmed sponsor-provided access/credits	Application development, interface editing, preview, and testing
RunPod	Serverless GPU platform and any confirmed sponsor-provided credits	Hosts Qwen3-4B-Instruct-2507 through vLLM for AI explanations and learning-plan generation

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

RunPod provides the current prototype's live Qwen inference, making it operationally important to AI-generated explanations and learning plans. Matching calculations and charts do not require GPU inference. Replit supports development and testing, while Vercel hosts the submitted website. Both sponsor platforms are technically replaceable with compatible alternatives, but replacement would require configuration and validation. Without an available inference provider, Faraway must report AI-generation failures rather than present a fabricated successful response.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Faraway collects account details, career interests, experience, skill ratings, learning preferences, and progress. Organisation users may enter names, positions, programmes/departments, and assessment ratings. CV inputs may contain additional personal information and should be reviewed before use.

Personal profiles and assessments are stored in Supabase with authenticated access and Row Level Security. Organisation records currently remain in account-scoped browser storage. AI requests send selected career interests, experience, skill ratings, learning preferences, and catalogue context through the backend to RunPod. The normalised personal-profile input excludes account email and name.

Server credentials remain in backend secrets. Frontend publishable keys rely on database access policies rather than secrecy. No automatic retention period has been established; records persist until removed through available controls or administrative processes. A complete deletion workflow and provider-log retention review remain outstanding.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The hardest risk is correctly matching dated occupational evidence to users' careers and skills. Sources differ in geography, coverage, and measurement; US projections are not Malaysian vacancy counts. Faraway uses source attribution, dates, reviewed mappings, and unavailable-data states, but mapping quality remains a limitation.

Moderate and rapid AI scenarios are planning assumptions, not validated forecasts. Self-reported skills and incomplete requirements can distort match scores. RunPod scale-to-zero cold starts, worker availability, timeouts, and limited credits can delay or prevent generation. Requests are bounded, and automatic retries are avoided.

AI can produce inaccurate, incomplete, or biased explanations. Structured-response validation and catalogue constraints reduce some errors but do not eliminate them. Multilingual text

requires further quality review. Fairness checks are limited and do not establish discrimination-free recommendations. Organisation records are browser-local and may be lost if browser storage is cleared. Further user testing and reliability checks are required.

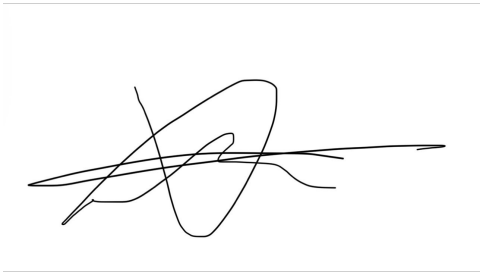
7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] 

[Date] 24-10-2026